Renaissance's Medallion Fund, which uses sophisticated algorithms to trade, has generated average annual returns of approximately 66% (before fees) since 1988—perhaps the most successful investment fund in history.

?? Fun Fact: Renaissance Technologies' founder, Jim Simons, was not a finance professional but a mathematician and code breaker who worked for the U.S. National Security Agency. His background in pattern recognition and statistical analysis proved more valuable for trading than traditional financial experience—highlighting how programming and mathematical skills can triumph in modern markets.

Algorithmic Trading Today

Today, algorithmic trading dominates financial markets across asset classes:

- **Equities**: More than 70% of U.S. equity trading volume is executed by algorithms
- **Foreign Exchange**: Approximately 80% of forex trading is algorithmic
- **Futures**: About 90% of futures trading is automated
- **Options**: More than 60% of options trading is algorithmic

The Flash Crash of May 6, 2010, demonstrated the power of algorithms in modern markets. In just 36 minutes, the Dow Jones Industrial Average dropped about 9% (roughly 1,000 points) before recovering most of those losses. Investigations later revealed that a combination of algorithmic trading and high-frequency trading magnified a large sell order, creating a cascade effect. This event highlighted both the power and the potential risks of algorithm-driven markets.